

TriFeaturizer

TriFeaturizer converts three kinds of chemical input: protein/peptide sequences, small-molecule SMILES, and inorganic materials (chemical formulas or CIF crystal structures) into complete, machine-learning-ready feature sets. This document is the reference for every feature it produces: what each one means, how many there are, and how they are delivered.

1 · How to use it

Workflow A - explore one item

Paste a single sequence, SMILES, or formula and choose one of three actions: Analyze (an explained table of the key descriptors), Charts (composition, hydropathy, drug-likeness, or lattice plots, exported as 600-dpi PNGs), or Full vector → CSV (the complete descriptor set for that one item).

Workflow B - process a whole dataset into Excel

Upload a CSV, Excel (.xlsx), or JSON file containing a column of inputs. TriFeaturizer detects the relevant column automatically by names such as smiles, sequence, or formula, and you may also type the column name explicitly. It then computes the full feature vector for every row and returns a single Excel workbook ready for modelling.

Input file: one row per item, with a column of SMILES, sequences, or formulas. Any other columns (identifiers, labels, target values) are ignored by the featurizer and can be re-joined to the output afterwards on the input column.

Output workbook: the first column echoes the input (input_SMILES, input_sequence, or input_formula); every remaining column is one descriptor. Rows that cannot be parsed are reported and skipped rather than aborting the run, and descriptors that are undefined for an otherwise-valid item appear as blanks. The file loads directly into pandas or scikit-learn.

Batch workbooks are named *part1_sequence_FULL_features.xlsx*, *part2_molecule_FULL_features.xlsx*, and *part3_composition_FULL_features.xlsx*.

2 · How many features per part

Part	Input	Engines	Analyze view	Full vector → Excel
1 · Sequence	residue / peptide / protein	Biopython + peptides.py	11 shown	≈ 108 (102 peptides.py + 6 Biopython)
2 · Molecule	SMILES	RDKit + mordredcommunity	13 shown	1613 (Mordred 2-D)
3 · Composition	chemical formula	pymatgen + matminer	14 shown	142 (132 Magpie + 4 valence-orbital stoichiometry) + 6
3 · Structure (CIF)	crystal file	pymatgen + matminer + DScibe	8 shown	8 structural + SOAP (≈ 390–1500, grows with #elements)

“Analyze view” = the explained values shown on screen. “Full vector → Excel” = the columns written to the workbook in batch mode (and to CSV for a single item). CIF structures are featurized one at a time rather than in batch.

3 · Sequence features

1.1 · Biopython (ProtParam)

Physico-chemical properties computed from the amino-acid sequence. The first eight appear in the Analyze view; the remainder are available from the same object and can be added to the export.

Feature	Units / range	What it measures
Length	residues	Number of amino acids in the chain.
Molecular weight	Da	Sum of residue masses minus one water per peptide bond.
Isoelectric point (pI)	pH 0–14	pH at which the molecule carries no net charge.
Net charge @ pH 7.0	± e	Net charge at neutral pH from residue pKa values.
GRAVY	≈ -4.5 ... +4.5	Grand average of hydropathy; positive = hydrophobic overall.
Aromaticity	0–1	Fraction of aromatic residues (Phe + Trp + Tyr).
Instability index	score	Dipeptide-based stability estimate; > 40 suggests instability in vitro.
Molar extinction coefficient	M ⁻¹ cm ⁻¹	Light absorption at 280 nm (reduced and cystine forms).
Flexibility	per-residue list	Vihinen flexibility profile along the chain (available).
Secondary-structure fraction	3 × 0–1	Predicted helix / turn / sheet fractions (available).
Amino-acid percent	20 × 0–1	Composition: fraction of each of the 20 residues (available).
Monoisotopic weight	Da	Exact mass using principal isotopes (available option).

1.2 · peptides.py - QSAR descriptor scales

peptides.py contributes 102 descriptors organised into 17 published QSAR “scale” families. Each family is a set of orthogonal components derived by multivariate analysis of amino-acid property tables; the components are sequence-averaged. The full set is what the Full vector → CSV button exports.

Family	n	Member labels	What it measures
Atchley factors	5	AF1, AF2, AF3, AF4, AF5	Five factors summarising polarity, secondary-structure propensity, molecular size/volume, codon diversity, and electrostatic charge (Atchley 2005).
BLOSUM indices	10	BLOSUM1, BLOSUM10, BLOSUM2, BLOSUM3, BLOSUM4, BLOSUM5, BLOSUM6, BLOSUM7, BLOSUM8, BLOSUM9	Principal-component scores from the BLOSUM62 substitution matrix, encoding substitution similarity (Georgiev 2009).
Cruciani properties	3	PP1, PP2, PP3	Scaled principal properties from molecular-interaction fields: polarity, hydrophobicity, H-bonding (Cruciani 2004).
FASGAI vectors	6	F1, F2, F3, F4, F5, F6	Factor-analysis scales: hydrophobicity, alpha-helix propensity, bulk/steric, composition, local flexibility, electronic (Liang 2007).
Kidera factors	10	KF1, KF10, KF2, KF3, KF4, KF5, KF6, KF7, KF8, KF9	Ten orthogonal factors from multivariate analysis of 188 physical properties (Kidera 1985).
MS-WHIM scores	3	MSWHIM1, MSWHIM2, MSWHIM3	Scores summarising 3-D electrostatic-potential surface properties (Zaliani 1999).
PCP descriptors	5	E1, E2, E3, E4, E5	Principal physico-chemical-property scores condensing charge, size and polarity.

Family	n	Member labels	What it measures
Physical descriptors	2	PD1, PD2	Two composite descriptors capturing residue volume and hydrophobicity.
PRIN components	3	PRIN1, PRIN2, PRIN3	Principal-component scores of amino-acid property indices.
ProtFP descriptors	8	ProtFP1, ProtFP2, ProtFP3, ProtFP4, ProtFP5, ProtFP6, ProtFP7, ProtFP8	Principal components of a large AAindex property set; a general protein fingerprint (van Westen 2013).
Sneath vectors	4	SV1, SV2, SV3, SV4	Principal components of Sneath's amino-acid similarity analysis (Sneath 1966).
ST-scales	8	ST1, ST2, ST3, ST4, ST5, ST6, ST7, ST8	Structural-topology scales derived from 827 amino-acid properties (Yang 2010).
SVGER descriptors	11	SVGER1, SVGER10, SVGER11, SVGER2, SVGER3, SVGER4, SVGER5, SVGER6, SVGER7, SVGER8, SVGER9	Eigenvalue-based descriptors from amino-acid property matrices.
T-scales	5	T1, T2, T3, T4, T5	Topological scales from PCA of structural descriptors (Tian 2007).
VHSE scales	8	VHSE1, VHSE2, VHSE3, VHSE4, VHSE5, VHSE6, VHSE7, VHSE8	Scores for hydrophobic (V), steric (H) and electronic (E) properties (Mei 2005).
VSTPV descriptors	6	VSTPV1, VSTPV2, VSTPV3, VSTPV4, VSTPV5, VSTPV6	Vectors of structural and physico-chemical variables.
Z-scales	5	Z1, Z2, Z3, Z4, Z5	Classic QSAR scales: lipophilicity, steric bulk/size, polarity, electronic effects (Sandberg 1998).

Total: 102 peptides.py descriptors. The notebook also adds 6 Biopython scalars (Length, MW, pI, GRAVY, Instability, Aromaticity) to the sequence CSV, giving ≈ 108 columns.

4 • Molecule features

2.1 • RDKit - curated descriptors (Analyze view)

Structure-aware descriptors plus a 2-D depiction, shown when you click Analyze.

Feature	Units / range	What it measures
Canonical SMILES	text	RDKit-normalised structure string.
Molecular formula	text	Elemental formula of the molecule.
Molecular weight	g/mol	Average molecular mass.
LogP (Crippen)	unitless	Calculated octanol–water partition coefficient; lipophilicity.
TPSA	Å ²	Topological polar surface area; a permeability / absorption proxy.
H-bond donors	count	Number of H-bond donors (Lipinski limit ≤ 5).
H-bond acceptors	count	Number of H-bond acceptors (Lipinski limit ≤ 10).
Rotatable bonds	count	Single non-ring bonds; molecular flexibility (Veber ≤ 10).
Aromatic rings	count	Number of aromatic ring systems.

Feature	Units / range	What it measures
Heavy atoms	count	Non-hydrogen atom count.
QED	0–1	Quantitative estimate of drug-likeness.
Lipinski violations	0–4	Number of rule-of-five breaches (0–1 typical for drugs).
Murcko scaffold	SMILES	Core ring framework with side chains removed.

2.2 · mordredcommunity - full descriptor vector

The Full vector → CSV button computes 1613 two-dimensional Mordred descriptors (3-D descriptors are disabled). They span 45 descriptor families, summarised below; every individual name is listed in Appendix A.

Descriptor family	n	What it measures
Autocorrelation	606	Moreau–Broto, Moran and Geary autocorrelations of atomic properties along the graph at several topological distances.
EState	316	Electrotopological-state indices encoding each atom's electronic environment and topology.
RingCount	138	Counts of rings by size, aromaticity, fusion and heteroatom content.
BaryszMatrix	104	Descriptors from the Barysz distance matrix weighted by atomic properties.
Chi	56	Kier–Hall connectivity (chi) indices of several orders and bond types.
MoeType	53	MOE-style VSA descriptors: surface area binned by LogP, molar refractivity and partial charge.
ExtendedTopochemicalAtom	45	ETA indices describing electronegativity, branching and composition.
InformationContent	42	Shannon information-content indices of atomic neighbourhoods.
BCUT	24	Burden-eigenvalue (BCUT) descriptors weighted by atomic properties.
PathCount	21	Counts of molecular paths of given lengths.
TopologicalCharge	21	Gálvez topological-charge indices from the adjacency and distance matrices.
WalkCount	21	Counts of self-returning and total walks of given lengths.
MolecularDistanceEdge	19	Distance-edge descriptors between carbon/heteroatom pairs.
AtomCount	17	Counts of atoms by element and hybridisation.
Constitutional	16	Constitutional sums and means of atomic properties.
DetourMatrix	14	Descriptors from the detour (longest-path) matrix.
AdjacencyMatrix	12	Spectral / eigenvalue descriptors of the adjacency matrix.
DistanceMatrix	12	Spectral / eigenvalue descriptors of the topological distance matrix.
MolecularId	12	Randić-type molecular identification numbers.
CarbonTypes	11	Counts of carbons by hybridisation and substitution pattern.
BondCount	9	Counts of bonds by order and type.
TopologicalIndex	4	Graph diameter, radius and related topological indices.
ZagrebIndex	4	First and second Zagreb indices and variants.

Descriptor family	n	What it measures
KappaShapeIndex	3	Kier kappa shape indices (orders 1–3).
ABCIndex	2	Atom-bond-connectivity indices.
AcidBase	2	Counts of acidic and basic groups.
Aromatic	2	Aromatic atom and bond counts.
CPSA	2	Charged partial-surface-area terms (2-D variant).
HydrogenBond	2	H-bond donor and acceptor counts.
Lipinski	2	Lipinski and Ghose-Crippen rule-of-five flags.
Polarizability	2	Atomic and bond polarizability sums.
RotatableBond	2	Rotatable-bond counts (including and excluding terminal bonds).
SLogP	2	Wildman–Crippen LogP and molar refractivity.
TopoPSA	2	Topological polar surface area (with and without S, P).
Weight	2	Average and exact (monoisotopic) molecular weight.
WienerIndex	2	Wiener path number and Wiener polarity number.
BalabanJ	1	Balaban's J connectivity index.
BertzCT	1	Bertz molecular-complexity index.
EccentricConnectivityIndex	1	Eccentric connectivity index.
FragmentComplexity	1	Fragment-based molecular-complexity score.
Framework	1	Fraction of the molecule belonging to its ring framework.
LogS	1	Estimated aqueous solubility (LogS).
McGowanVolume	1	McGowan characteristic molecular volume.
VdwVolumeABC	1	Van der Waals volume (atom-and-bond-contribution method).
VertexAdjacencyInformation	1	Vertex-adjacency information content.

5 · Material features

3a · Composition (formula)

matminer ElementProperty - Magpie preset (132 features)

Each of the 22 elemental properties below is reduced to 6 composition-level statistics, giving $22 \times 6 = 132$ features. The statistics, computed across the elements of the formula weighted by stoichiometric fraction, are: minimum, maximum, range, mean, avg_dev (mean absolute deviation) and mode.

Elemental property	What it measures
AtomicWeight	Standard atomic weight (amu).
Column	Group (column) in the periodic table.
CovalentRadius	Covalent radius (pm).
Electronegativity	Pauling electronegativity.
GSbandgap	DFT ground-state band gap (eV).

Elemental property	What it measures
GSmagmom	DFT ground-state magnetic moment (μB).
GSvolume_pa	DFT ground-state volume per atom ($\text{\AA}^3$).
MeltingT	Elemental melting temperature (K).
MendeleeevNumber	Mendeleeev number — a chemically-ordered element index.
NUnfilled	Total number of unfilled valence states.
NValence	Total number of valence electrons.
NdUnfilled	Number of unfilled d valence states.
NdValence	Number of valence electrons in the d shell.
NfUnfilled	Number of unfilled f valence states.
NfValence	Number of valence electrons in the f shell.
NpUnfilled	Number of unfilled p valence states.
NpValence	Number of valence electrons in the p shell.
NsUnfilled	Number of unfilled s valence states.
NsValence	Number of valence electrons in the s shell.
Number	Atomic number (proton count).
Row	Period (row) in the periodic table.
SpaceGroupNumber	Space-group number of the elemental ground-state structure.

Full names follow the pattern “MagpieData <statistic> <Property>”, e.g. MagpieData mean Electronegativity. All 132 are listed in Appendix B.

matminer ValenceOrbital - valence fractions (4 features)

Feature	What it measures
frac s valence electrons	Fraction of the composition's valence electrons residing in the s orbital.
frac p valence electrons	Fraction of the composition's valence electrons residing in the p orbital.
frac d valence electrons	Fraction of the composition's valence electrons residing in the d orbital.
frac f valence electrons	Fraction of the composition's valence electrons residing in the f orbital.

matminer Stoichiometry - composition p-norms (6 features)

Feature	What it measures
0-norm	Number of distinct elements in the formula.
2-norm	Euclidean norm of the fractional-composition vector.
3-norm	Third-order p-norm of the composition vector.
5-norm	Fifth-order p-norm (weights majority elements more).
7-norm	Seventh-order p-norm of the composition vector.
10-norm	Tenth-order p-norm (approaches the dominant element fraction).

3b · Structure (CIF crystal file)

matminer DensityFeatures (3 features)

Feature	Units	What it measures
density	g/cm ³	Crystal mass density.
vpa	Å ³ /atom	Volume per atom in the unit cell.
packing fraction	0–1	Fraction of cell volume filled by atomic spheres.

matminer GlobalSymmetryFeatures (5 features)

Feature	What it measures
spacegroup_num	International space-group number (1–230).
crystal_system	Crystal system name (cubic, tetragonal, ...).
crystal_system_int	Integer encoding of the crystal system.
is Centrosymmetric	Whether the structure has an inversion centre (0/1).
n_symmetry_ops	Number of symmetry operations in the space group.

Dscribe SOAP - local-environment fingerprint (optional)

SOAP (Smooth Overlap of Atomic Positions) encodes each atom's local geometry as a power spectrum; the notebook averages over all atoms to give one fixed-length vector per structure. Settings used: $r_{\text{cut}} = 5.0 \text{ \AA}$, $n_{\text{max}} = 6$, $l_{\text{max}} = 4$, `periodic = True`.

The vector length grows with the number of distinct elements present, so it is not a fixed number:

Distinct elements (example)	SOAP vector length
2 (e.g. NaCl)	390
3 (e.g. BaTiO ₃)	855
4 (e.g. LiFePO ₄)	1500

Names are emitted as SOAP₀ ... SOAP_(n-1).

Appendix A · All 1613 Mordred (2-D) feature names

Grouped by family. These are the exact column names written to the molecule full-vector CSV.

Autocorrelation (606)

ATS0dv, ATS1dv, ATS2dv, ATS3dv, ATS4dv, ATS5dv, ATS6dv, ATS7dv, ATS8dv, ATS0d, ATS1d, ATS2d, ATS3d, ATS4d, ATS5d, ATS6d, ATS7d, ATS8d, ATS0s, ATS1s, ATS2s, ATS3s, ATS4s, ATS5s, ATS6s, ATS7s, ATS8s, ATS0Z, ATS1Z, ATS2Z, ATS3Z, ATS4Z, ATS5Z, ATS6Z, ATS7Z, ATS8Z, ATS0m, ATS1m, ATS2m, ATS3m, ATS4m, ATS5m, ATS6m, ATS7m, ATS8m, ATS0v, ATS1v, ATS2v, ATS3v, ATS4v, ATS5v, ATS6v, ATS7v, ATS8v, ATS0se, ATS1se, ATS2se, ATS3se, ATS4se, ATS5se, ATS6se, ATS7se, ATS8se, ATS0pe, ATS1pe, ATS2pe, ATS3pe, ATS4pe, ATS5pe, ATS6pe, ATS7pe, ATS8pe, ATS0are, ATS1are, ATS2are, ATS3are, ATS4are, ATS5are, ATS6are, ATS7are, ATS8are, ATS0p, ATS1p, ATS2p, ATS3p, ATS4p, ATS5p, ATS6p, ATS7p, ATS8p, ATS0i, ATS1i, ATS2i, ATS3i, ATS4i, ATS5i, ATS6i, ATS7i, ATS8i, AATS0dv, AATS1dv, AATS2dv, AATS3dv, AATS4dv, AATS5dv, AATS6dv, AATS7dv, AATS8dv, AATS0d, AATS1d, AATS2d, AATS3d, AATS4d, AATS5d, AATS6d, AATS7d, AATS8d, AATS0s, AATS1s, AATS2s, AATS3s, AATS4s, AATS5s, AATS6s, AATS7s, AATS8s, AATS0Z, AATS1Z, AATS2Z, AATS3Z, AATS4Z, AATS5Z, AATS6Z, AATS7Z, AATS8Z, AATS0m, AATS1m, AATS2m, AATS3m, AATS4m, AATS5m, AATS6m, AATS7m, AATS8m, AATS0v, AATS1v, AATS2v, AATS3v, AATS4v, AATS5v, AATS6v, AATS7v, AATS8v, AATS0se, AATS1se, AATS2se, AATS3se, AATS4se, AATS5se, AATS6se, AATS7se, AATS8se, AATS0pe, AATS1pe, AATS2pe, AATS3pe, AATS4pe, AATS5pe, AATS6pe, AATS7pe, AATS8pe, AATS0are, AATS1are, AATS2are, AATS3are, AATS4are, AATS5are, AATS6are, AATS7are, AATS8are, AATS0p, AATS1p, AATS2p, AATS3p, AATS4p, AATS5p, AATS6p, AATS7p, AATS8p, AATS0i, AATS1i, AATS2i, AATS3i, AATS4i, AATS5i, AATS6i, AATS7i, AATS8i, ATSC0c, ATSC1c, ATSC2c, ATSC3c, ATSC4c, ATSC5c, ATSC6c, ATSC7c, ATSC8c, ATSC0dv, ATSC1dv, ATSC2dv, ATSC3dv, ATSC4dv, ATSC5dv, ATSC6dv, ATSC7dv, ATSC8dv, ATSC0d, ATSC1d, ATSC2d, ATSC3d, ATSC4d, ATSC5d, ATSC6d, ATSC7d, ATSC8d, ATSC0s, ATSC1s, ATSC2s, ATSC3s, ATSC4s, ATSC5s, ATSC6s, ATSC7s, ATSC8s, ATSC0Z, ATSC1Z, ATSC2Z, ATSC3Z, ATSC4Z, ATSC5Z, ATSC6Z, ATSC7Z, ATSC8Z, ATSC0m, ATSC1m, ATSC2m, ATSC3m, ATSC4m, ATSC5m, ATSC6m, ATSC7m, ATSC8m, ATSC0v, ATSC1v, ATSC2v, ATSC3v, ATSC4v, ATSC5v, ATSC6v, ATSC7v, ATSC8v, ATSC0se, ATSC1se, ATSC2se, ATSC3se, ATSC4se, ATSC5se, ATSC6se, ATSC7se, ATSC8se, ATSC0pe, ATSC1pe, ATSC2pe, ATSC3pe, ATSC4pe, ATSC5pe, ATSC6pe, ATSC7pe, ATSC8pe, ATSC0are, ATSC1are, ATSC2are, ATSC3are, ATSC4are, ATSC5are, ATSC6are, ATSC7are, ATSC8are, ATSC0p, ATSC1p, ATSC2p, ATSC3p, ATSC4p, ATSC5p, ATSC6p, ATSC7p, ATSC8p, ATSC0i, ATSC1i, ATSC2i, ATSC3i, ATSC4i, ATSC5i, ATSC6i, ATSC7i, ATSC8i, AATSC0c, AATSC1c, AATSC2c, AATSC3c, AATSC4c, AATSC5c, AATSC6c, AATSC7c, AATSC8c, AATSC0dv, AATSC1dv, AATSC2dv, AATSC3dv, AATSC4dv, AATSC5dv, AATSC6dv, AATSC7dv, AATSC8dv, AATSC0d, AATSC1d, AATSC2d, AATSC3d, AATSC4d, AATSC5d, AATSC6d, AATSC7d, AATSC8d, AATSC0s, AATSC1s, AATSC2s, AATSC3s, AATSC4s, AATSC5s, AATSC6s, AATSC7s, AATSC8s, AATSC0Z, AATSC1Z, AATSC2Z, AATSC3Z, AATSC4Z, AATSC5Z, AATSC6Z, AATSC7Z, AATSC8Z, AATSC0m, AATSC1m, AATSC2m, AATSC3m, AATSC4m, AATSC5m, AATSC6m, AATSC7m, AATSC8m, AATSC0v, AATSC1v, AATSC2v, AATSC3v, AATSC4v, AATSC5v, AATSC6v, AATSC7v, AATSC8v, AATSC0se, AATSC1se, AATSC2se, AATSC3se, AATSC4se, AATSC5se, AATSC6se, AATSC7se, AATSC8se, AATSC0pe, AATSC1pe, AATSC2pe, AATSC3pe, AATSC4pe, AATSC5pe, AATSC6pe, AATSC7pe, AATSC8pe, AATSC0are, AATSC1are, AATSC2are, AATSC3are, AATSC4are, AATSC5are, AATSC6are, AATSC7are, AATSC8are, AATSC0p, AATSC1p, AATSC2p, AATSC3p, AATSC4p, AATSC5p, AATSC6p, AATSC7p, AATSC8p, AATSC0i, AATSC1i, AATSC2i, AATSC3i, AATSC4i, AATSC5i, AATSC6i, AATSC7i, AATSC8i, MATS1c, MATS2c, MATS3c, MATS4c, MATS5c, MATS6c, MATS7c, MATS8c, MATS1dv, MATS2dv, MATS3dv, MATS4dv, MATS5dv, MATS6dv, MATS7dv, MATS8dv, MATS1d, MATS2d, MATS3d, MATS4d, MATS5d, MATS6d, MATS7d, MATS8d, MATS1s, MATS2s, MATS3s, MATS4s, MATS5s, MATS6s, MATS7s, MATS8s, MATS1Z, MATS2Z, MATS3Z, MATS4Z, MATS5Z, MATS6Z, MATS7Z, MATS8Z, MATS1m, MATS2m, MATS3m, MATS4m, MATS5m, MATS6m, MATS7m, MATS8m, MATS1v, MATS2v, MATS3v, MATS4v, MATS5v, MATS6v, MATS7v, MATS8v, MATS1se, MATS2se, MATS3se, MATS4se, MATS5se, MATS6se, MATS7se, MATS8se, MATS1pe, MATS2pe, MATS3pe, MATS4pe, MATS5pe, MATS6pe, MATS7pe, MATS8pe, MATS1are, MATS2are, MATS3are, MATS4are, MATS5are, MATS6are, MATS7are, MATS8are, MATS1p, MATS2p, MATS3p, MATS4p, MATS5p, MATS6p, MATS7p, MATS8p, MATS1i, MATS2i, MATS3i, MATS4i, MATS5i, MATS6i, MATS7i, MATS8i, GATS1c, GATS2c, GATS3c, GATS4c, GATS5c, GATS6c, GATS7c, GATS8c, GATS1dv, GATS2dv, GATS3dv, GATS4dv, GATS5dv, GATS6dv, GATS7dv, GATS8dv, GATS1d, GATS2d, GATS3d, GATS4d, GATS5d, GATS6d, GATS7d, GATS8d, GATS1s, GATS2s, GATS3s, GATS4s, GATS5s, GATS6s, GATS7s, GATS8s, GATS1Z, GATS2Z, GATS3Z, GATS4Z, GATS5Z, GATS6Z, GATS7Z, GATS8Z, GATS1m, GATS2m, GATS3m, GATS4m, GATS5m, GATS6m, GATS7m, GATS8m,

GATS1v, GATS2v, GATS3v, GATS4v, GATS5v, GATS6v, GATS7v, GATS8v, GATS1se, GATS2se, GATS3se, GATS4se, GATS5se, GATS6se, GATS7se, GATS8se, GATS1pe, GATS2pe, GATS3pe, GATS4pe, GATS5pe, GATS6pe, GATS7pe, GATS8pe, GATS1are, GATS2are, GATS3are, GATS4are, GATS5are, GATS6are, GATS7are, GATS8are, GATS1p, GATS2p, GATS3p, GATS4p, GATS5p, GATS6p, GATS7p, GATS8p, GATS1i, GATS2i, GATS3i, GATS4i, GATS5i, GATS6i, GATS7i, GATS8i

EState (316)

NsLi, NssBe, NssssBe, NssBH, NsssB, NssssB, NsCH3, NdCH2, NssCH2, NtCH, NdsCH, NaaCH, NssssCH, NddC, NtsC, NdssC, NaasC, NaaaC, NssssC, NsNH3, NsNH2, NssNH2, NdNH, NssNH, NaaNH, NtN, NssssNH, NdsN, NaaN, NssssN, NddsN, NaasN, NssssN, NsOH, NdO, NssO, NaaO, NsF, NsSiH3, NssSiH2, NssssSiH, NssssSi, NsPH2, NssPH, NssP, NdssP, NssssP, NsSH, NdS, NssS, NaaS, NdssS, NddssS, NsCl, NsGeH3, NssGeH2, NssssGeH, NssssGe, NsAsH2, NssAsH, NssssAs, NsssdAs, NssssAs, NsSeH, NdSe, NssSe, NaaSe, NdssSe, NddssSe, NsBr, NsSnH3, NssSnH2, NssssSnH, NssssSn, NsI, NsPbH3, NssPbH2, NssssPbH, NssssPb, SsLi, SssBe, SssssBe, SssBH, SssssB, SssssB, SsCH3, SdCH2, SssCH2, StCH, SdsCH, SaaCH, SssssCH, SddC, StsC, SdssC, SaasC, SaaaC, SssssC, SsNH3, SsNH2, SssNH2, SdNH, SssNH, SaaNH, StN, SssssNH, SdsN, SaaN, SssssN, SddsN, SaasN, SssssN, SsOH, SdO, SssO, SaaO, SsF, SsSiH3, SssSiH2, SssssSiH, SssssSi, SsPH2, SssPH, SssP, SdssP, SssssP, SsSH, SdS, SssS, SaaS, SdssS, SddssS, SsCl, SsGeH3, SssGeH2, SssssGeH, SssssGe, SsAsH2, SssAsH, SssssAs, SsssdAs, SssssAs, SsSeH, SdSe, SssSe, SaaSe, SdssSe, SddssSe, SsBr, SsSnH3, SssSnH2, SssssSnH, SssssSn, Ssl, SsPbH3, SssPbH2, SssssPbH, SssssPb, MAXsLi, MAXssBe, MAXssssBe, MAXssBH, MAXssB, MAXssssB, MAXsCH3, MAXdCH2, MAXssCH2, MAXtCH, MAXdsCH, MAXaaCH, MAXsssCH, MAXddC, MAXtsC, MAXdssC, MAXaasC, MAXaaaC, MAXssssC, MAXsNH3, MAXsNH2, MAXssNH2, MAXdNH, MAXssNH, MAXaaNH, MAXtN, MAXsssNH, MAXdsN, MAXaaN, MAXssN, MAXddsN, MAXaasN, MAXssssN, MAXsOH, MAXdO, MAXssO, MAXaaO, MAXsF, MAXsSiH3, MAXssSiH2, MAXsssSiH, MAXssssSi, MAXsPH2, MAXssPH, MAXssP, MAXdssP, MAXssssP, MAXsSH, MAXdS, MAXssS, MAXaaS, MAXdssS, MAXddssS, MAXsCl, MAXsGeH3, MAXssGeH2, MAXsssGeH, MAXssssGe, MAXsAsH2, MAXssAsH, MAXsssAs, MAXsssdAs, MAXssssAs, MAXsSeH, MAXdSe, MAXssSe, MAXaaSe, MAXdssSe, MAXddssSe, MAXsBr, MAXsSnH3, MAXssSnH2, MAXssssSnH, MAXssssSn, MAXsI, MAXsPbH3, MAXssPbH2, MAXsssPbH, MAXssssPb, MINsLi, MINssBe, MINssssBe, MINssBH, MINssB, MINssssB, MINsCH3, MINdCH2, MINssCH2, MINtCH, MINdsCH, MINaaCH, MINssCH, MINddC, MINtsC, MINdssC, MINaasC, MINaaaC, MINssssC, MINsNH3, MINsNH2, MINssNH2, MINdNH, MINssNH, MINaaNH, MINtN, MINssNH, MINdsN, MINaaN, MINssN, MINddsN, MINaasN, MINssssN, MINsOH, MINdO, MINssO, MINaaO, MINsF, MINsSiH3, MINssSiH2, MINssSiH, MINssssSi, MINsPH2, MINssPH, MINssP, MINdssP, MINssssP, MINsSH, MINdS, MINssS, MINaaS, MINdssS, MINddssS, MINsCl, MINsGeH3, MINssGeH2, MINssssGeH, MINssssGe, MINsAsH2, MINssAsH, MINssssAs, MINsssdAs, MINssssAs, MINsSeH, MINdSe, MINssSe, MINaaSe, MINdssSe, MINddssSe, MINsBr, MINsSnH3, MINssSnH2, MINssssSnH, MINssssSn, MINsI, MINsPbH3, MINssPbH2, MINssPbH, MINssssPb

RingCount (138)

nRing, n3Ring, n4Ring, n5Ring, n6Ring, n7Ring, n8Ring, n9Ring, n10Ring, n11Ring, n12Ring, nG12Ring, nHRing, n3HRing, n4HRing, n5HRing, n6HRing, n7HRing, n8HRing, n9HRing, n10HRing, n11HRing, n12HRing, nG12HRing, naRing, n3aRing, n4aRing, n5aRing, n6aRing, n7aRing, n8aRing, n9aRing, n10aRing, n11aRing, n12aRing, nG12aRing, naHRing, n3aHRing, n4aHRing, n5aHRing, n6aHRing, n7aHRing, n8aHRing, n9aHRing, n10aHRing, n11aHRing, n12aHRing, nG12aHRing, nARing, n3ARing, n4ARing, n5ARing, n6ARing, n7ARing, n8ARing, n9ARing, n10ARing, n11ARing, n12ARing, nG12ARing, nAHRing, n3AHRing, n4AHRing, n5AHRing, n6AHRing, n7AHRing, n8AHRing, n9AHRing, n10AHRing, n11AHRing, n12AHRing, nG12AHRing, nFRing, n4FRing, n5FRing, n6FRing, n7FRing, n8FRing, n9FRing, n10FRing, n11FRing, n12FRing, nG12FRing, nFHRing, n4FHRing, n5FHRing, n6FHRing, n7FHRing, n8FHRing, n9FHRing, n10FHRing, n11FHRing, n12FHRing, nG12FHRing, nFaRing, n4FaRing, n5FaRing, n6FaRing, n7FaRing, n8FaRing, n9FaRing, n10FaRing, n11FaRing, n12FaRing, nG12FaRing, nFaHRing, n4FaHRing, n5FaHRing, n6FaHRing, n7FaHRing, n8FaHRing, n9FaHRing, n10FaHRing, n11FaHRing, n12FaHRing, nG12FaHRing, nFARing, n4FARing, n5FARing, n6FARing, n7FARing, n8FARing, n9FARing, n10FARing, n11FARing, n12FARing, nG12FARing, nFAHRing, n4FAHRing, n5FAHRing, n6FAHRing, n7FAHRing, n8FAHRing, n9FAHRing, n10FAHRing, n11FAHRing, n12FAHRing

BaryszMatrix (104)

SpAbs_DzZ, SpMax_DzZ, SpDiam_DzZ, SpAD_DzZ, SpMAD_DzZ, LogEE_DzZ, SM1_DzZ, VE1_DzZ, VE2_DzZ, VE3_DzZ, VR1_DzZ, VR2_DzZ, VR3_DzZ, SpAbs_Dzm, SpMax_Dzm, SpDiam_Dzm, SpAD_Dzm, SpMAD_Dzm, LogEE_Dzm, SM1_Dzm, VE1_Dzm, VE2_Dzm, VE3_Dzm, VR1_Dzm, VR2_Dzm, VR3_Dzm, SpAbs_Dzv,

SpMax_Dzv, SpDiam_Dzv, SpAD_Dzv, SpMAD_Dzv, LogEE_Dzv, SM1_Dzv, VE1_Dzv, VE2_Dzv, VE3_Dzv, VR1_Dzv, VR2_Dzv, VR3_Dzv, SpAbs_Dzse, SpMax_Dzse, SpDiam_Dzse, SpAD_Dzse, SpMAD_Dzse, LogEE_Dzse, SM1_Dzse, VE1_Dzse, VE2_Dzse, VE3_Dzse, VR1_Dzse, VR2_Dzse, VR3_Dzse, SpAbs_Dzpe, SpMax_Dzpe, SpDiam_Dzpe, SpAD_Dzpe, SpMAD_Dzpe, LogEE_Dzpe, SM1_Dzpe, VE1_Dzpe, VE2_Dzpe, VE3_Dzpe, VR1_Dzpe, VR2_Dzpe, VR3_Dzpe, SpAbs_Dzare, SpMax_Dzare, SpDiam_Dzare, SpAD_Dzare, SpMAD_Dzare, LogEE_Dzare, SM1_Dzare, VE1_Dzare, VE2_Dzare, VE3_Dzare, VR1_Dzare, VR2_Dzare, VR3_Dzare, SpAbs_Dzp, SpMax_Dzp, SpDiam_Dzp, SpAD_Dzp, SpMAD_Dzp, LogEE_Dzp, SM1_Dzp, VE1_Dzp, VE2_Dzp, VE3_Dzp, VR1_Dzp, VR2_Dzp, VR3_Dzp, SpAbs_Dzi, SpMax_Dzi, SpDiam_Dzi, SpAD_Dzi, SpMAD_Dzi, LogEE_Dzi, SM1_Dzi, VE1_Dzi, VE2_Dzi, VE3_Dzi, VR1_Dzi, VR2_Dzi, VR3_Dzi

Chi (56)

Xch-3d, Xch-4d, Xch-5d, Xch-6d, Xch-7d, Xch-3dv, Xch-4dv, Xch-5dv, Xch-6dv, Xch-7dv, Xc-3d, Xc-4d, Xc-5d, Xc-6d, Xc-3dv, Xc-4dv, Xc-5dv, Xc-6dv, Xpc-4d, Xpc-5d, Xpc-6d, Xpc-4dv, Xpc-5dv, Xpc-6dv, Xp-0d, Xp-1d, Xp-2d, Xp-3d, Xp-4d, Xp-5d, Xp-6d, Xp-7d, AXp-0d, AXp-1d, AXp-2d, AXp-3d, AXp-4d, AXp-5d, AXp-6d, AXp-7d, Xp-0dv, Xp-1dv, Xp-2dv, Xp-3dv, Xp-4dv, Xp-5dv, Xp-6dv, Xp-7dv, AXp-0dv, AXp-1dv, AXp-2dv, AXp-3dv, AXp-4dv, AXp-5dv, AXp-6dv, AXp-7dv

MoeType (53)

LabuteASA, PEOE_VSA1, PEOE_VSA2, PEOE_VSA3, PEOE_VSA4, PEOE_VSA5, PEOE_VSA6, PEOE_VSA7, PEOE_VSA8, PEOE_VSA9, PEOE_VSA10, PEOE_VSA11, PEOE_VSA12, PEOE_VSA13, SMR_VSA1, SMR_VSA2, SMR_VSA3, SMR_VSA4, SMR_VSA5, SMR_VSA6, SMR_VSA7, SMR_VSA8, SMR_VSA9, SlogP_VSA1, SlogP_VSA2, SlogP_VSA3, SlogP_VSA4, SlogP_VSA5, SlogP_VSA6, SlogP_VSA7, SlogP_VSA8, SlogP_VSA9, SlogP_VSA10, SlogP_VSA11, EState_VSA1, EState_VSA2, EState_VSA3, EState_VSA4, EState_VSA5, EState_VSA6, EState_VSA7, EState_VSA8, EState_VSA9, EState_VSA10, VSA_EState1, VSA_EState2, VSA_EState3, VSA_EState4, VSA_EState5, VSA_EState6, VSA_EState7, VSA_EState8, VSA_EState9

ExtendedTopochemicalAtom (45)

ETA_alpha, AETA_alpha, ETA_shape_p, ETA_shape_y, ETA_shape_x, ETA_beta, AETA_beta, ETA_beta_s, AETA_beta_s, ETA_beta_ns, AETA_beta_ns, ETA_beta_ns_d, AETA_beta_ns_d, ETA_eta, AETA_eta, ETA_eta_L, AETA_eta_L, ETA_eta_R, AETA_eta_R, ETA_eta_RL, AETA_eta_RL, ETA_eta_F, AETA_eta_F, ETA_eta_FL, AETA_eta_FL, ETA_eta_B, AETA_eta_B, ETA_eta_BR, AETA_eta_BR, ETA_dAlpha_A, ETA_dAlpha_B, ETA_epsilon_1, ETA_epsilon_2, ETA_epsilon_3, ETA_epsilon_4, ETA_epsilon_5, ETA_dEpsilon_A, ETA_dEpsilon_B, ETA_dEpsilon_C, ETA_dEpsilon_D, ETA_dBeta, AETA_dBeta, ETA_psi_1, ETA_dPsi_A, ETA_dPsi_B

InformationContent (42)

IC0, IC1, IC2, IC3, IC4, IC5, TIC0, TIC1, TIC2, TIC3, TIC4, TIC5, SIC0, SIC1, SIC2, SIC3, SIC4, SIC5, BIC0, BIC1, BIC2, BIC3, BIC4, BIC5, CIC0, CIC1, CIC2, CIC3, CIC4, CIC5, MIC0, MIC1, MIC2, MIC3, MIC4, MIC5, ZMIC0, ZMIC1, ZMIC2, ZMIC3, ZMIC4, ZMIC5

BCUT (24)

BCUTc-1h, BCUTc-1l, BCUTdv-1h, BCUTdv-1l, BCUTd-1h, BCUTd-1l, BCUTs-1h, BCUTs-1l, BCUTZ-1h, BCUTZ-1l, BCUTm-1h, BCUTm-1l, BCUTv-1h, BCUTv-1l, BCUTse-1h, BCUTse-1l, BCUTpe-1h, BCUTpe-1l, BCUTare-1h, BCUTare-1l, BCUTp-1h, BCUTp-1l, BCUTi-1h, BCUTi-1l

PathCount (21)

MPC2, MPC3, MPC4, MPC5, MPC6, MPC7, MPC8, MPC9, MPC10, TMPC10, piPC1, piPC2, piPC3, piPC4, piPC5, piPC6, piPC7, piPC8, piPC9, piPC10, TpiPC10

TopologicalCharge (21)

GGI1, GGI2, GGI3, GGI4, GGI5, GGI6, GGI7, GGI8, GGI9, GGI10, JGI1, JGI2, JGI3, JGI4, JGI5, JGI6, JGI7, JGI8, JGI9, JGI10, JGT10

WalkCount (21)

MWC01, MWC02, MWC03, MWC04, MWC05, MWC06, MWC07, MWC08, MWC09, MWC10, TMWC10, SRW02, SRW03, SRW04, SRW05, SRW06, SRW07, SRW08, SRW09, SRW10, TSRW10

MolecularDistanceEdge (19)

MDEC-11, MDEC-12, MDEC-13, MDEC-14, MDEC-22, MDEC-23, MDEC-24, MDEC-33, MDEC-34, MDEC-44, MDEO-11, MDEO-12, MDEO-22, MDEN-11, MDEN-12, MDEN-13, MDEN-22, MDEN-23, MDEN-33

AtomCount (17)

nAtom, nHeavyAtom, nSpiro, nBridgehead, nHetero, nH, nB, nC, nN, nO, nS, nP, nF, nCl, nBr, nI, nX

Constitutional (16)

SZ, Sm, Sv, Sse, Spe, Sare, Sp, Si, MZ, Mm, Mv, Mse, Mpe, Mare, Mp, Mi

DetourMatrix (14)

SpAbs_Dt, SpMax_Dt, SpDiam_Dt, SpAD_Dt, SpMAD_Dt, LogEE_Dt, SM1_Dt, VE1_Dt, VE2_Dt, VE3_Dt, VR1_Dt, VR2_Dt, VR3_Dt, DetourIndex

AdjacencyMatrix (12)

SpAbs_A, SpMax_A, SpDiam_A, SpAD_A, SpMAD_A, LogEE_A, VE1_A, VE2_A, VE3_A, VR1_A, VR2_A, VR3_A

DistanceMatrix (12)

SpAbs_D, SpMax_D, SpDiam_D, SpAD_D, SpMAD_D, LogEE_D, VE1_D, VE2_D, VE3_D, VR1_D, VR2_D, VR3_D

MolecularId (12)

MID, AMID, MID_h, AMID_h, MID_C, AMID_C, MID_N, AMID_N, MID_O, AMID_O, MID_X, AMID_X

CarbonTypes (11)

C1SP1, C2SP1, C1SP2, C2SP2, C3SP2, C1SP3, C2SP3, C3SP3, C4SP3, HybRatio, FCSP3

BondCount (9)

nBonds, nBondsO, nBondsS, nBondsD, nBondsT, nBondsA, nBondsM, nBondsKS, nBondsKD

TopologicalIndex (4)

Diameter, Radius, TopoShapeIndex, PetitjeanIndex

ZagrebIndex (4)

Zagreb1, Zagreb2, mZagreb1, mZagreb2

KappaShapelIndex (3)

Kier1, Kier2, Kier3

ABCIndex (2)

ABC, ABCGG

AcidBase (2)

nAcid, nBase

Aromatic (2)

nAromAtom, nAromBond

CPSA (2)

RNCG, RPCG

HydrogenBond (2)

nHBAcc, nHBDon

Lipinski (2)

Lipinski, GhoseFilter

Polarizability (2)

apol, bpol

RotatableBond (2)

nRot, RotRatio

SLogP (2)

SLogP, SMR

TopoPSA (2)

TopoPSA(NO), TopoPSA

Weight (2)

MW, AMW

WienerIndex (2)

WPath, WPol

BalabanJ (1)

BalabanJ

BertzCT (1)

BertzCT

EccentricConnectivityIndex (1)

ECIndex

FragmentComplexity (1)

fragCpx

Framework (1)

fMF

LogS (1)

FilterItLogS

McGowanVolume (1)

VMcGowan

VdwVolumeABC (1)

Vabc

VertexAdjacencyInformation (1)

VAdjMat

Appendix B · All 132 Magpie composition feature names

Pattern: MagpieData <statistic> <Property>.

MagpieData minimum Number, MagpieData maximum Number, MagpieData range Number, MagpieData mean Number, MagpieData avg_dev Number, MagpieData mode Number, MagpieData minimum MendelevNumber, MagpieData maximum MendelevNumber, MagpieData range MendelevNumber, MagpieData mean MendelevNumber, MagpieData avg_dev MendelevNumber, MagpieData mode MendelevNumber, MagpieData minimum AtomicWeight, MagpieData maximum AtomicWeight, MagpieData range AtomicWeight, MagpieData mean AtomicWeight, MagpieData avg_dev AtomicWeight, MagpieData mode AtomicWeight, MagpieData minimum MeltingT, MagpieData maximum MeltingT, MagpieData range MeltingT, MagpieData mean MeltingT, MagpieData avg_dev MeltingT, MagpieData mode MeltingT, MagpieData minimum Column, MagpieData maximum Column, MagpieData range Column, MagpieData mean Column, MagpieData avg_dev Column, MagpieData mode Column, MagpieData minimum Row, MagpieData maximum Row, MagpieData range Row, MagpieData mean Row, MagpieData avg_dev Row, MagpieData mode Row, MagpieData minimum CovalentRadius, MagpieData maximum CovalentRadius, MagpieData range CovalentRadius, MagpieData mean CovalentRadius, MagpieData avg_dev CovalentRadius, MagpieData mode CovalentRadius, MagpieData minimum Electronegativity, MagpieData maximum Electronegativity, MagpieData range Electronegativity, MagpieData mean Electronegativity, MagpieData avg_dev Electronegativity, MagpieData mode Electronegativity, MagpieData minimum NsValence, MagpieData maximum NsValence, MagpieData range NsValence, MagpieData mean NsValence, MagpieData avg_dev NsValence, MagpieData mode NsValence, MagpieData minimum NpValence, MagpieData maximum NpValence, MagpieData range NpValence, MagpieData mean NpValence, MagpieData avg_dev NpValence, MagpieData mode NpValence, MagpieData minimum NdValence, MagpieData maximum NdValence, MagpieData range NdValence, MagpieData mean NdValence, MagpieData avg_dev NdValence, MagpieData mode NdValence, MagpieData minimum NfValence, MagpieData maximum NfValence, MagpieData range NfValence, MagpieData mean NfValence, MagpieData avg_dev NfValence, MagpieData mode NfValence, MagpieData minimum NValence, MagpieData maximum NValence, MagpieData range NValence, MagpieData mean NValence, MagpieData avg_dev NValence, MagpieData mode NValence, MagpieData minimum NsUnfilled, MagpieData maximum NsUnfilled, MagpieData range NsUnfilled, MagpieData mean NsUnfilled, MagpieData avg_dev NsUnfilled, MagpieData mode NsUnfilled, MagpieData minimum NpUnfilled, MagpieData maximum NpUnfilled, MagpieData range NpUnfilled, MagpieData mean NpUnfilled, MagpieData avg_dev NpUnfilled, MagpieData mode NpUnfilled, MagpieData minimum NdUnfilled, MagpieData maximum NdUnfilled, MagpieData range NdUnfilled, MagpieData mean NdUnfilled, MagpieData avg_dev NdUnfilled, MagpieData mode NdUnfilled, MagpieData minimum NfUnfilled, MagpieData maximum NfUnfilled, MagpieData range NfUnfilled, MagpieData mean NfUnfilled, MagpieData avg_dev NfUnfilled, MagpieData mode NfUnfilled, MagpieData minimum NUnfilled, MagpieData maximum NUnfilled, MagpieData range NUnfilled, MagpieData mean NUnfilled, MagpieData avg_dev NUnfilled, MagpieData mode NUnfilled, MagpieData minimum GSvolume_pa, MagpieData maximum GSvolume_pa, MagpieData range GSvolume_pa, MagpieData mean GSvolume_pa, MagpieData avg_dev GSvolume_pa, MagpieData mode GSvolume_pa, MagpieData minimum GSbandgap, MagpieData maximum GSbandgap, MagpieData range GSbandgap, MagpieData mean GSbandgap, MagpieData avg_dev GSbandgap, MagpieData mode GSbandgap, MagpieData minimum GSmagmom, MagpieData maximum GSmagmom, MagpieData range GSmagmom, MagpieData mean GSmagmom, MagpieData avg_dev GSmagmom, MagpieData mode GSmagmom, MagpieData minimum SpaceGroupNumber, MagpieData maximum SpaceGroupNumber, MagpieData range SpaceGroupNumber, MagpieData mean SpaceGroupNumber, MagpieData avg_dev SpaceGroupNumber, MagpieData mode SpaceGroupNumber